

Backbone



Conv P1/2

Conv P2/4

Cf2 (P2)

Conv P3/8

Cf2 (P3)

Conv P4/16

Cf2 (P4)

Conv P5/32

Cf2 (P5)

SPPF

Head/Neck

P3_fused

Conv 3×3

Conv 1×1

SGCBlock (P3)

Conv 1×1

Upsample

P4_fused

Conv 3×3

Conv 1×1

SGCBlock (P4)

Conv 1×1

Upsample

P5_refined

AMRF

Conv 1×1

ScaleMapHead

AvgPool 2×2 /2

AvgPool 2×2 /2

Scale-map branch

DETECT HEAD

Detect (P3_fused)
small

Detect (P4_fused)
medium

Detect (P5_refined)
large